# A microfluidic system for label-free and high-throughput magnetic separation of nanoparticles

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## ABSTRACT

A microfluidic system for label-free and high-throughput magnetic separation of nanoparticles is reported, and the performance of the microfluidic system is studied under different flow rates. An ultrahigh magnetic field is generated in the microfluidic channel by fabricating an on-chip integrated magnetic micro-pole array, hereby establishing a negative magnetophoretic technique to separate nanoparticles in this device. The optimal flow rate of 2.5 μL/min is obtained from both simulation and experimental results..

**KEYWORDS:** Microfluidic chip, Label-free, Magnetic separation, Flow rate, Nanoscale

## INTRODUCTION

Label-free magnetic separation based on negative magnetophoresis has the advantages of simple operation and no damage to biological samples compared with magnetic labeling manner [1-3]. In recent years, the resolution of label-free magnetic separation has been evolving from micron-scale to nano-scale [4,5]. However, as the influence of Brownian motion on nanoparticles becomes more and more obvious, the influence of multiple factors on the separation effect are more complicated. In this paper, a high-throughput microfluidic system for label-free magnetic separating of nanoparticles is proposed and systematically studied.

## THEORY

The chip design is shown in Figure 1A. The particle samples mixed with 0.01% ferrofluid are first focused by the sheath fluid on both sides of the channel, and then they experience a negative magnetophoretic force that is proportional to the particle's volume, which makes large particles enter outlet A, and small particles enter outlet B. The magnetic field distribution (simulated by COMSOL Multiphysics) of the separation area is shown in Figure 1B. The magnetic field intensity near the magnetic pole arrays reaches 2.4 T, and the magnetic field gradient exceeds 10000 T/m, which can provide great negative magnetophoretic force for nanoparticle separation.

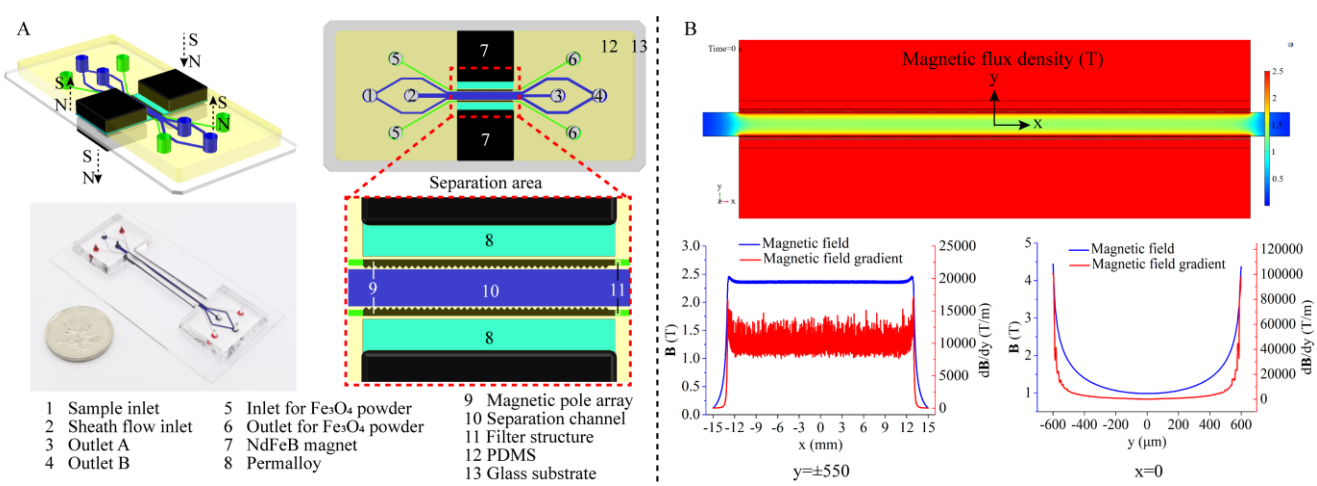


*Figure 1: A) Schematic illustration and photograph of the microfluidic system; B) Simulation on the magnetic field.*

## RESULTS AND DISCUSSION

The fluorescent particles with sizes of 200 nm (~35000 particles/μL) and 1 μm (~3500 particles/μL) are used in separation experiments, the flow rate (sample inlet) is set to 1-3 μL/min by a micro-injection pump, and the flow sheath ratio is 1:5. The results recorded by a fluorescence microscope is shown in Figure 2A.

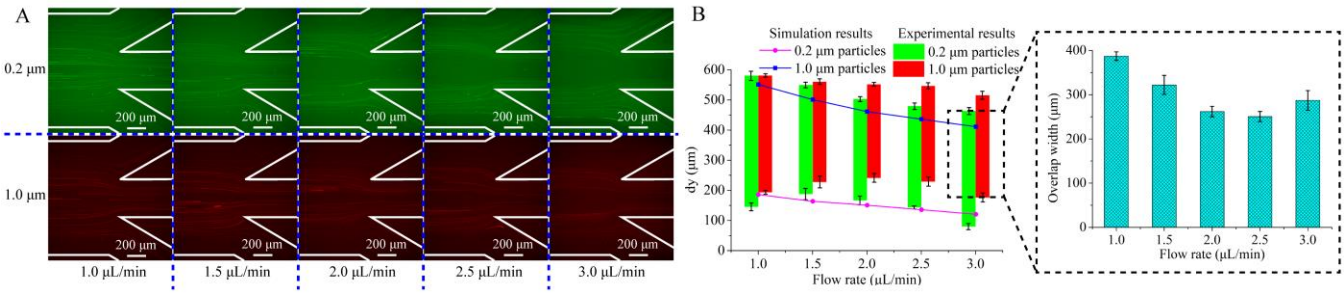


*Figure 2: Separation of 200 nm and 1 μm particles at different flow rates.*

With the increase of the flow rate, the trajectory of the 200 nm and 1 μm particles gradually moves from the middle to the sides of the microchannel. Then we calculate the particle distribution along the y-axis at the outlets (Figure 2B), and the results show that there is an overlap between the two kinds of particles along the y-axis, which means that the two particles are not completely separated. Thereafter, we obtain the optimal flow rate of 2.5 μL/min, at which the overlap width of the two kinds of particles is the shortest. The results also show that when the flow rate is too slow (1 μL/min), the impact of the hydrodynamic drag force will be weakened, and the Brownian motion effect will be more obvious, leading to a more scattered distribution of the particles at outlet.

## CONCLUSION

In this paper, a microfluidic chip for nanoparticle separation based on negative magnetophoresis has been presented, the effect of flow rate on the separation is studied via simulation and experiments. The results show that the optimal flow rate for the proposed microfluidic device to separate 200 nm and 1 μm particles is 2.5 μL/min, at which the overlap width of the two kinds of particles is the shortest.


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